College of Business Education

MODULE NAME	**COMPUTER SYSTEMS ADMINISTRATION – CLASS NOTES**
MODULE CODE	**ITU07401**
LEVEL	**BIT 2**
ACADEMIC YEAR	**2025/2026**
SEMESTER	**2**

- Access the Share from a Linux desktop
  - Open File Explorer and enter login path

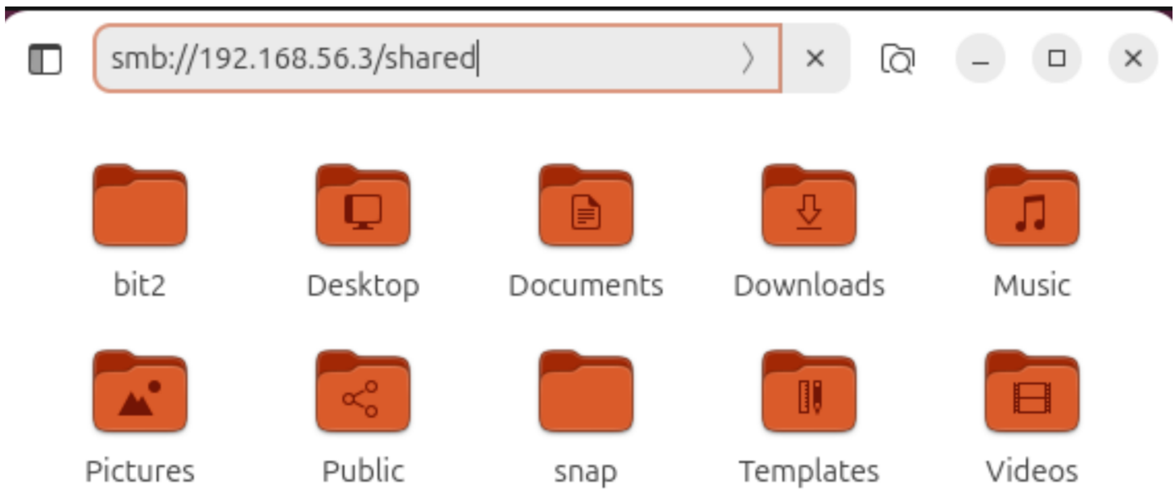


Figure Lab3-11: Exploring the file-dhcp-server from Windows Explorer in Ubuntu desktop

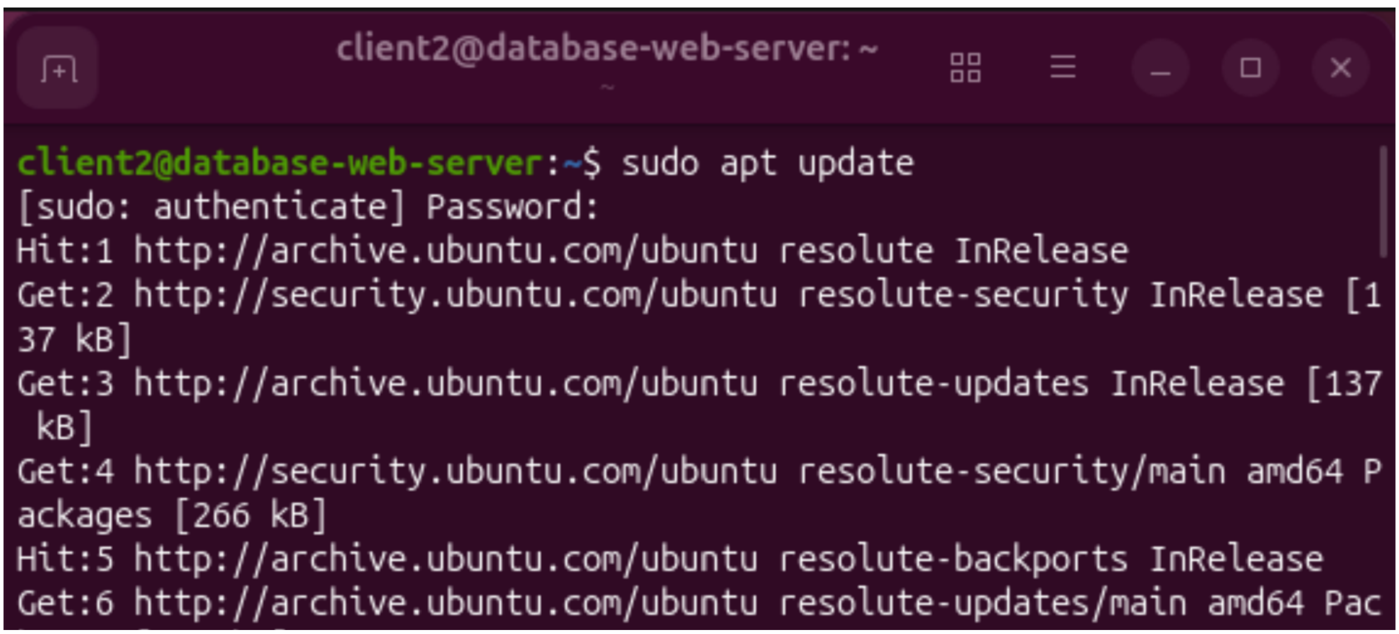
9 Configuring Web-Server

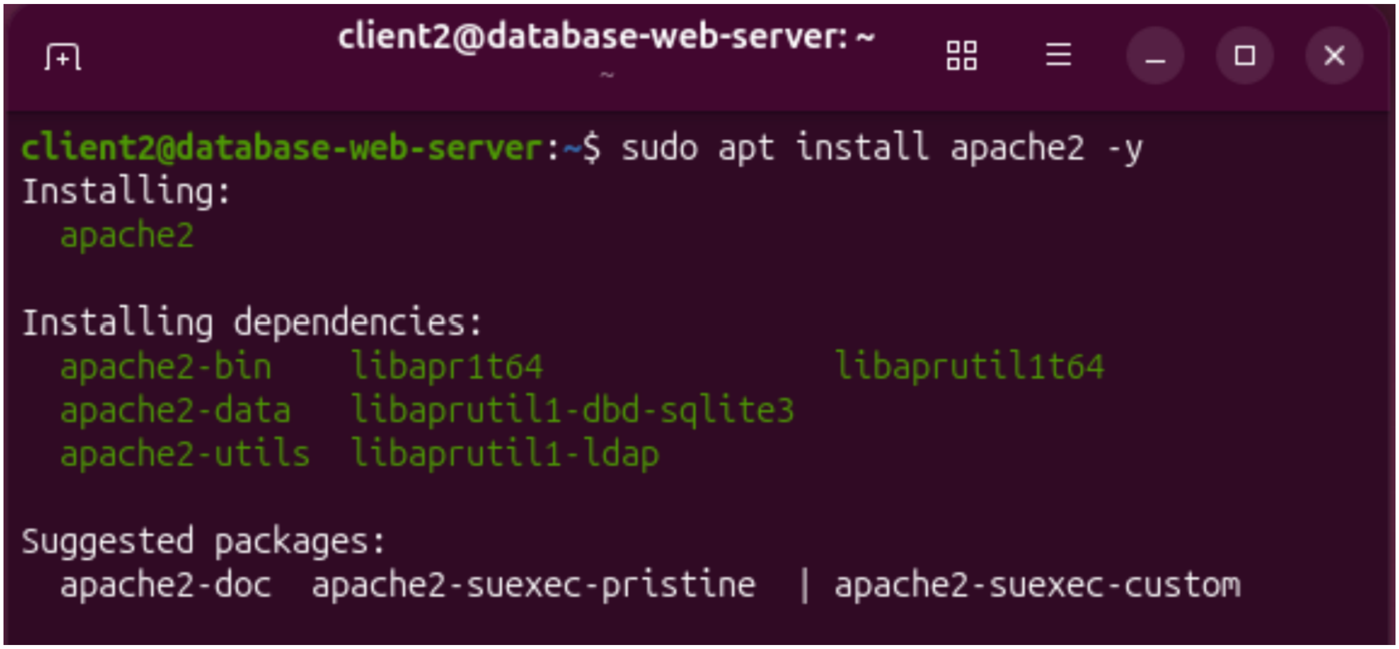
Configuring a web server on Ubuntu depends on whether you want to use **Apache** or **Nginx**. Here are the steps to configure the Apache web server.

i) Configure the web server on database-web-server

- Install Apache

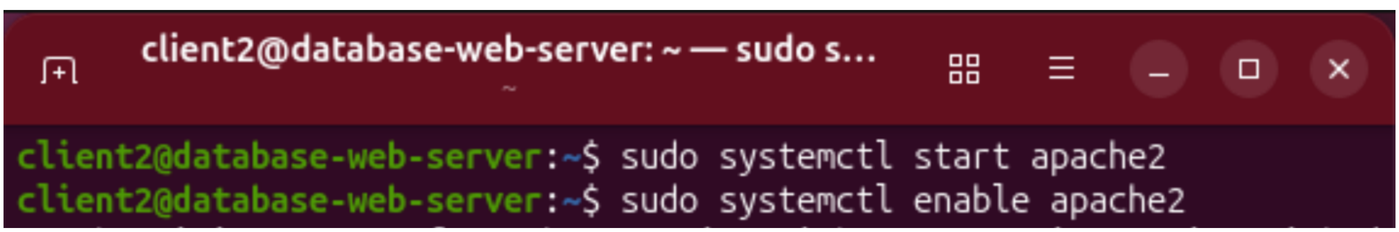
```
sudo apt update
sudo apt install apache2 -y
```





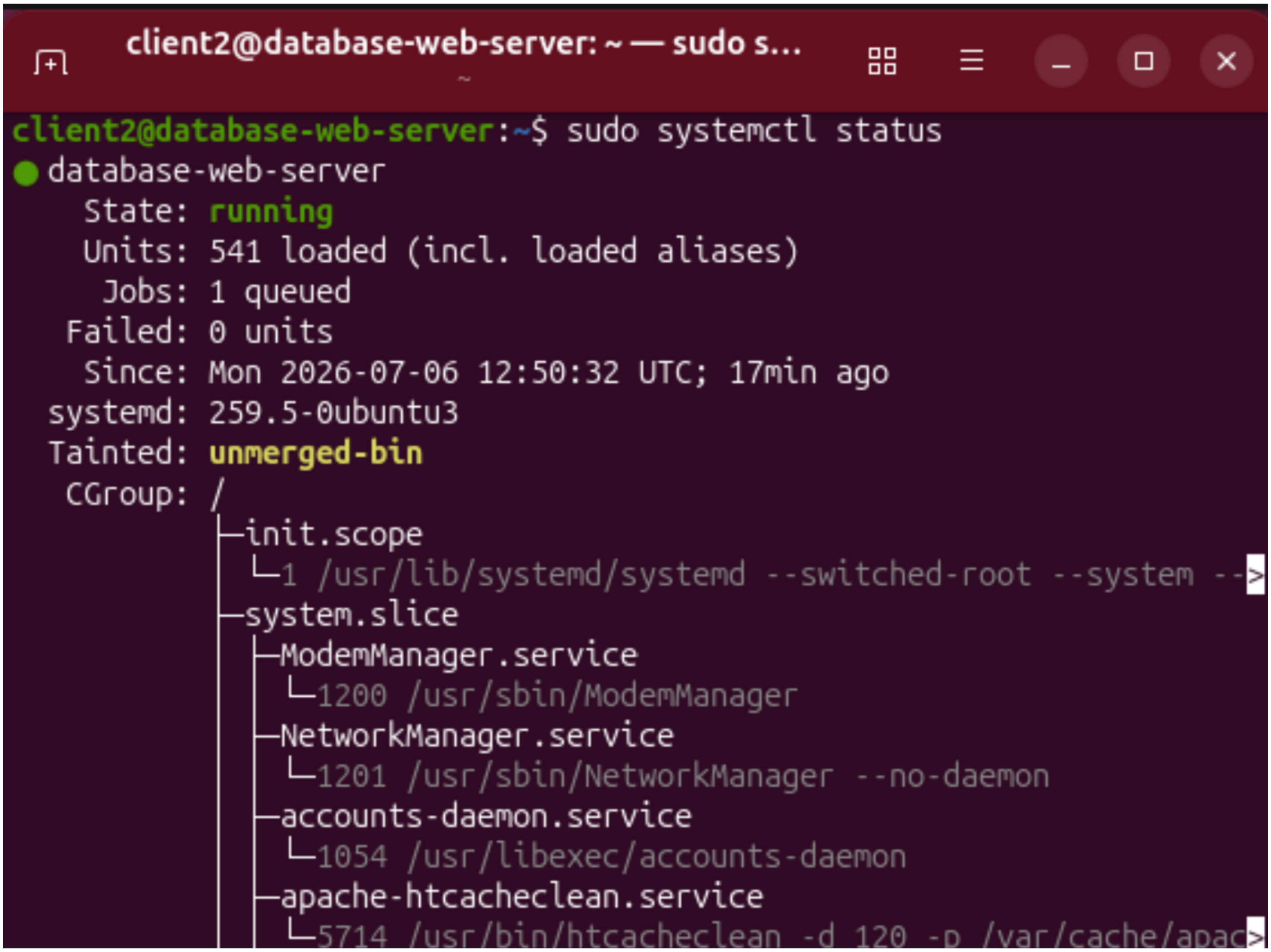
- ➢ Start and enable Apache

```
sudo systemctl start apache2
sudo systemctl enable apache2
```



- ➢ Check its status

```
sudo systemctl status apache2
```



- Allow HTTP and HTTPS through the firewall

```
sudo ufw allow 'Apache Full'
sudo ufw enable
sudo ufw status
```

```
client2@database-web-server:~$ sudo ufw allow "Apache Full"
Rules updated
Rules updated (v6)
client2@database-web-server:~$ sudo ufw enable
Firewall is active and enabled on system startup
client2@database-web-server:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
Apache Full                ALLOW       Anywhere
Apache Full (v6)           ALLOW       Anywhere (v6)

client2@database-web-server:~$
```

- Observe if the test page exists in the /var/www/html folder on the web server
  - Navigating to the folder

    ```
    cd /var/www/html
    ```

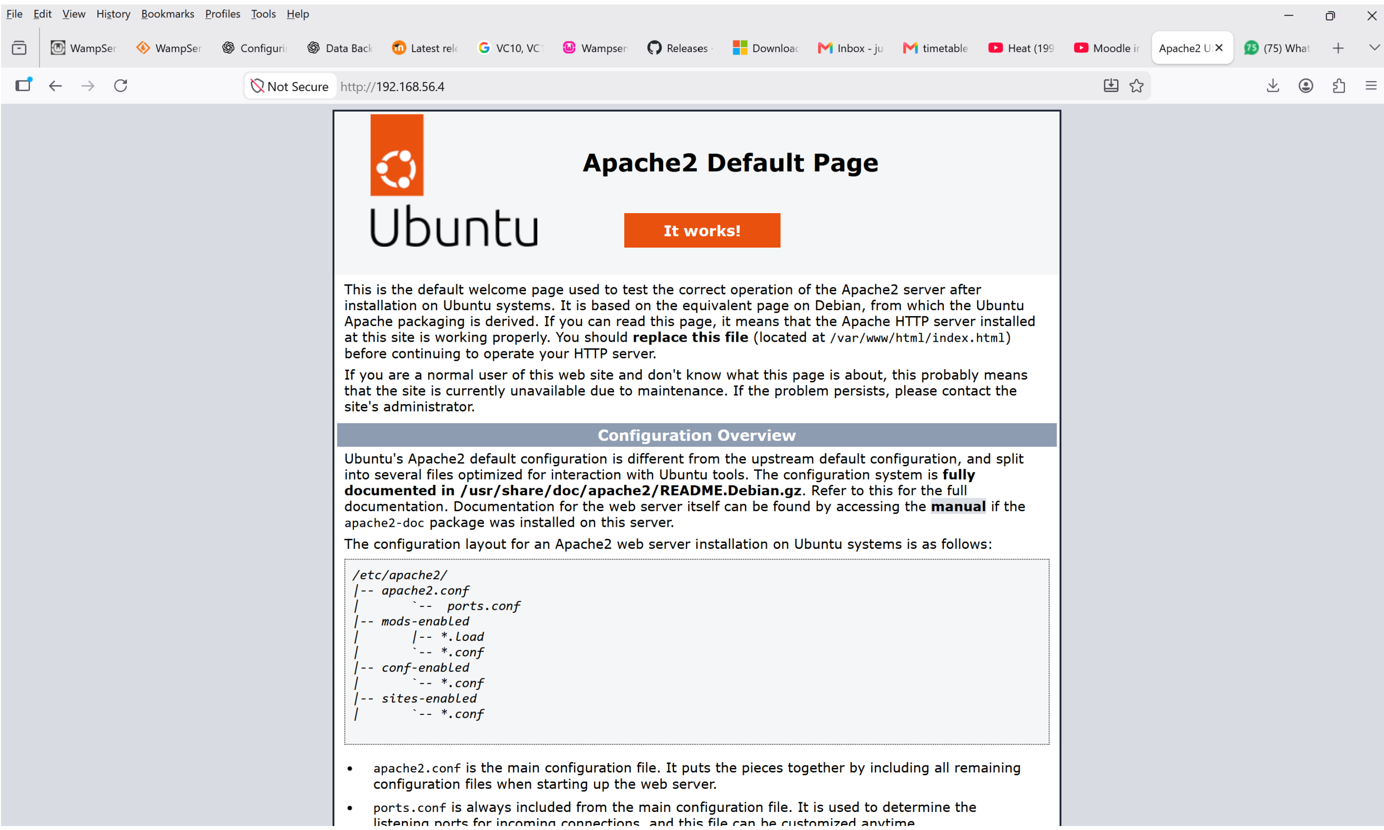
  - Use the ls command to view the folder contents, and you will see the default web page named **index.php** in the folder. If the default page does not exist, you can upload any **.html** document using the FileZilla tool installed on the client machine.

    ```
    ls
    ```

```
client2@database-web-server:/var/www/html$ ls
index.html
```

- Test the server
  - Open the web browser on the client machine (Host Computer) and type the IP address of the **database-web-server ->** The default web page will display

192.168.52.4



File Edit View History Bookmarks Profiles Tools Help
Not Secure http://192.168.56.4
Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled
| |-- *.load
| `-- *.conf
|-- conf-enabled
| `-- *.conf
|-- sites-enabled
| `-- *.conf
apache2.conf is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
ports.conf is always included from the main configuration file. It is used to determine the

10Data Backup and Recovery in Computer Systems Administration

Introduction

Modern organizations depend heavily on digital data for daily functions, decision-making, communication, and service provision. Losing data can lead to financial loss, operational disruptions, legal issues, and reputational harm. Therefore, systems administrators must develop robust backup and recovery plans to ensure business continuity and disaster preparedness. Data backup involves creating secure copies of data for restoration if the original becomes inaccessible, while data recovery is the process of restoring this information after failures, attacks, or accidental deletions. An effective backup strategy includes not only creating copies but also regularly testing and validating them.

10.1 Types and Backup Procedures

Definition of Data Backup

A data backup is a duplicate copy of information stored separately from the original data source to recover the information if the original data is lost, damaged, corrupted, or destroyed.

Backups protect organizations against:

- Hardware failures such as hard disk crashes.
- Human errors, including accidental deletion of files.
- Malware and ransomware attacks.
- Software failures and corruption.
- Power outages and system crashes.
- Natural disasters such as floods, fires, and earthquakes.
- Theft or sabotage.

Systems administrators design backup solutions based on the organization's recovery requirements, available storage, and acceptable downtime.

10.2 Types of Backups

1. Full Backup

A full backup creates a complete copy of all selected files, folders, applications, and system data every time the backup process runs.

- **Example:**

If an organization has 500 GB of data, a full backup copies all 500 GB during every backup cycle.

Advantages

i) Simplifies restoration because only one backup set is required.

ii) Provides the highest level of protection.

iii) Makes disaster recovery faster.

Disadvantages

i) Requires large amounts of storage space.

ii) Consumes significant network bandwidth.

iii) Takes longer to complete than other backup methods.

Typical Usage

- Weekly or monthly backups.
- Initial backup before implementing incremental or differential backups.

2. Incremental Backup

An incremental backup copies only the files that have changed since the last backup operation, whether that backup was full or incremental.

Example:

- Monday: Full backup.
- Tuesday: Backup only files changed since Monday.
- Wednesday: Backup only files changed since Tuesday.

Advantages

i) Uses minimal storage space.

ii) Reduces backup time significantly.

iii) Suitable for daily backups.

Disadvantages

i) Restoration is more complex.

ii) Recovery requires the full backup plus every incremental backup made afterward.

iii) A damaged incremental file may affect restoration.

Typical Usage

- Daily or hourly backups in active environments.

3. Differential Backup

A differential backup copies all files that have changed since the last full backup.

Example:

- Monday: Full backup.
- Tuesday: Backup changes since Monday.
- Wednesday: Backup all changes since Monday again.

Advantages

- Faster restoration than incremental backups.
- Requires fewer backup files during recovery.

Disadvantages

- Uses more storage than incremental backups.
- Backup size grows larger each day until the next full backup.

Typical Usage

- Organizations requiring faster recovery times.

4. Mirror Backup

A mirror backup creates an exact copy of the source files without compression or modification.

Advantages

i) Files are immediately accessible.

ii) Recovery is extremely fast.

iii) No extraction software is required.

Disadvantages

i) Accidental deletion on the source may also delete files from the mirror.

ii) Does not provide version history.

Typical Usage

- Real-time replication systems.
- High-availability environments.

5. Image Backup

An image backup creates a complete snapshot of an entire system, including:

- Operating system
- Applications
- Configuration settings
- User files
- Boot information

Advantages

i) Enables full system restoration.

ii) Reduces downtime after hardware failure.

Disadvantages

i) Requires substantial storage space.

ii) Backup creation takes longer.

Typical Usage

- Server backups.
- Critical production systems.

6. Cloud Backup

Cloud backup stores information on remote servers managed by cloud providers.

Advantages

i) Accessible from any location.

ii) Provides geographic redundancy.

iii) Supports automatic backup scheduling.

Disadvantages

i) Internet speed affects backup and restoration performance.

ii) Subscription costs may increase over time.

Typical Usage

- Disaster recovery planning.
- Remote work environments.

10.3 Backup Procedures Used by Systems Administrators

Step 1: Identify Critical Data

Administrators determine which information is essential to business operations, such as:

- Databases
- Financial records
- Customer information
- System configurations
- Application files

Step 2: Determine Recovery Objectives

Two important measures are defined:

Recovery Point Objective (RPO)

The maximum acceptable amount of data loss is measured in time.

Example:
If the RPO is 4 hours, backups must occur at least every 4 hours.

Recovery Time Objective (RTO)

The maximum acceptable downtime before services are restored.

Example:
An RTO of two hours means systems must be operational within two hours after failure.

Step 3: Select Backup Method

Administrators choose between:

- Full backups
- Incremental backups
- Differential backups
- Hybrid approaches

Step 4: Schedule Backups

Typical schedules include:

- Daily incremental backups
- Weekly full backups
- Monthly archival backups

Step 5: Monitor Backup Jobs

Administrators review:

- Backup logs
- Error reports

- Storage usage
- Completion notifications

Step 6: Test Restoration

Regular restoration testing confirms that backups are usable.

Step 7: Document Procedures

Documentation includes:

- Backup schedules
- Storage locations
- Recovery procedures
- Administrator responsibilities

10.4 On-Site and Off-Site Backups

On-Site Backups

On-site backups are stored within the organization's premises using:

- External hard drives
- Tape libraries
- Network Attached Storage (NAS)
- Storage Area Networks (SAN)

Advantages

i) **Faster Recovery**

Since data is stored locally, restoration speeds are high.

ii) **Lower Internet Dependency**

Recovery does not depend on network connectivity.

iii) **Greater Administrative Control**

Organizations maintain direct control over storage devices.

iv) Reduced Recurring Costs

After initial investment, ongoing costs are relatively low.

Disadvantages

i) Vulnerability to Local Disasters

Fire, flooding, theft, or power surges can destroy both original data and backups.

ii) Physical Security Risks

Unauthorized access may compromise backup data.

iii) Maintenance Costs

Hardware replacement and maintenance are required.

Off-Site Backups

Off-site backups are stored in geographically separate locations.

Examples include:

- Cloud storage services
- Secondary data centers
- Remote branch offices

Advantages

i) Disaster Protection

Backups remain safe if the primary site is destroyed.

ii) Geographic Redundancy

Protects against regional disasters.

iii) Scalability

Storage can easily expand as data grows.

iv) Remote Accessibility

Administrators can recover data from any location.

Disadvantages

i) **Slower Recovery**

Large restorations may take considerable time.

ii) **Internet Dependence**

Recovery performance depends on network bandwidth.

iii) **Ongoing Costs**

Cloud subscriptions and bandwidth charges can be expensive.

iv) **Security Concerns**

Data must be encrypted to prevent unauthorized access.

Comparison between On-site and Off-site Backup

Factor	**On-Site Backup**	**Off-Site Backup**
Recovery Speed	Very fast	Moderate to slow
Disaster Protection	Low	High
Initial Cost	High	Low to moderate
Ongoing Cost	Low	Higher subscription costs
Accessibility	Local access only	Global access
Scalability	Limited by hardware	Highly scalable